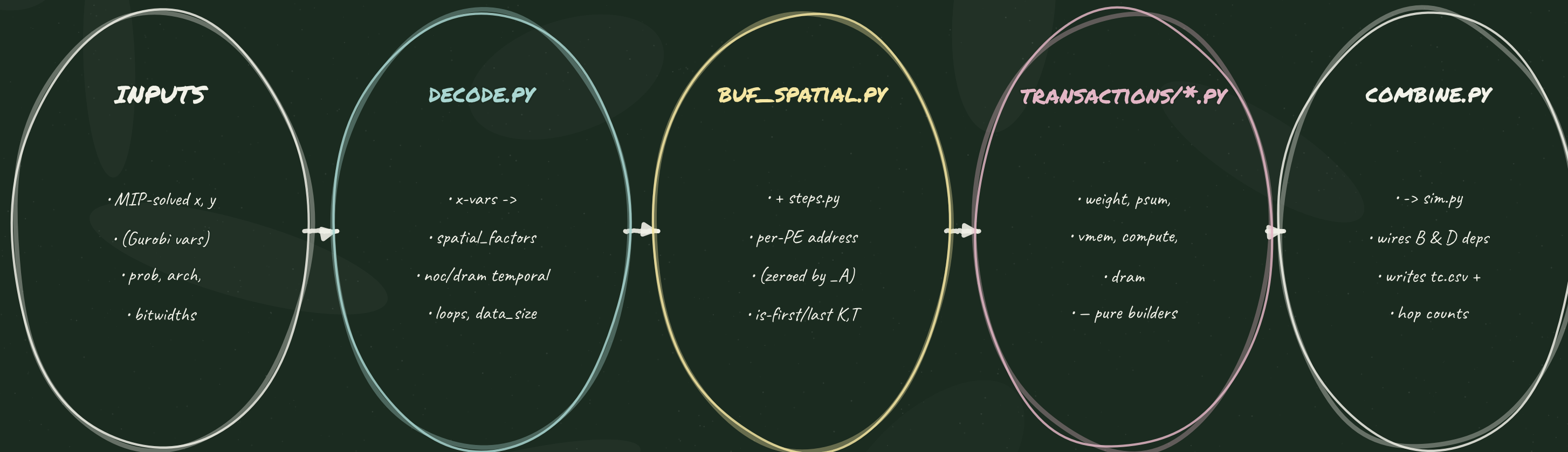


NOC SIMULATOR: MIP SOLUTION -> TRAFFIC CSV



each stage consumes only the previous stage's output - strictly left to right

ONE NOC TEMPORAL STEP: THE DEPENDENCY CHAIN

LOAD WEIGHT

- GB $\rightarrow$ all nodes (multicast)
- dep: prior weight send (S)
- dep: MAC 2 steps ago (B)

LOAD PSUM

- GB $\rightarrow$ K_max, skip if first-K
- dep: psum store -2 steps (B)
- dep: same-T store -T_noc (D)

MAC $\rightarrow$ K-CHAIN $\rightarrow$ STORE PSUM

- MAC COUNT, all nodes parallel
- K-chain: K=0 $\rightarrow$... $\rightarrow$ K_max, serial
- store K_max $\rightarrow$ GB, skip if last-K

only if last-K

LOAD VMEM

- GB $\rightarrow$ T_min, skip if first-T
- dep: vmem_ring[-2] (B)
- dep: vmem_ring[-1] (D)

LIF $\rightarrow$ T-CHAIN $\rightarrow$ STORE VMEM

- LIF COUNT, all nodes parallel
- T-chain: T=0 $\rightarrow$... $\rightarrow$ T_max, serial
- store T_max $\rightarrow$ GB, skip if last-T

LEGEND

B = buffer dep (offset -2)

D = data dep (must commit first)